

Electron-coupling-to-waveguide

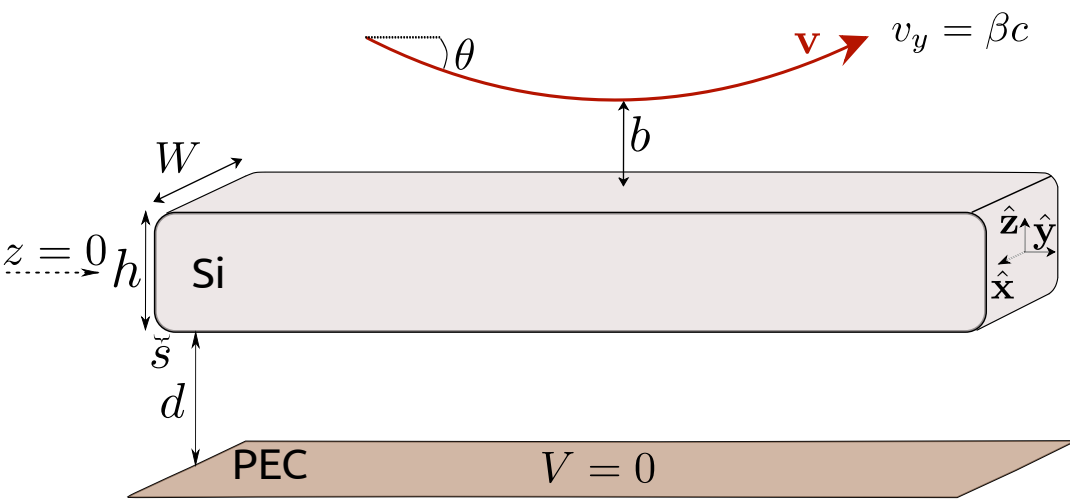
$E_{\perp} = E_{\text{electron}}\theta^2$ marked in dashed black line

$\Delta E_{\perp} = 2E_{\text{electron}}\theta\Delta\theta$ marked in grey

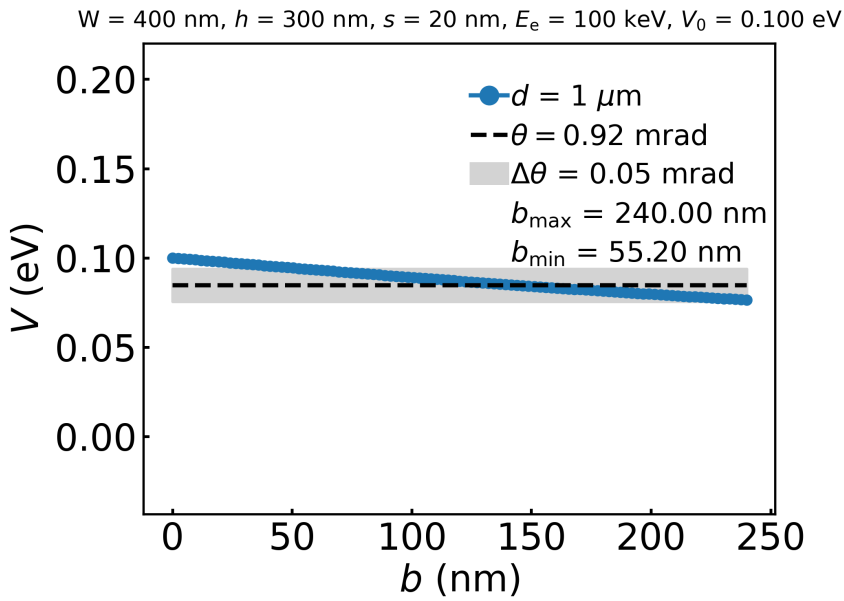
$W = 400 \text{ nm} \quad h = 300 \mu\text{m}$

$E_e = 100 \text{ keV}$

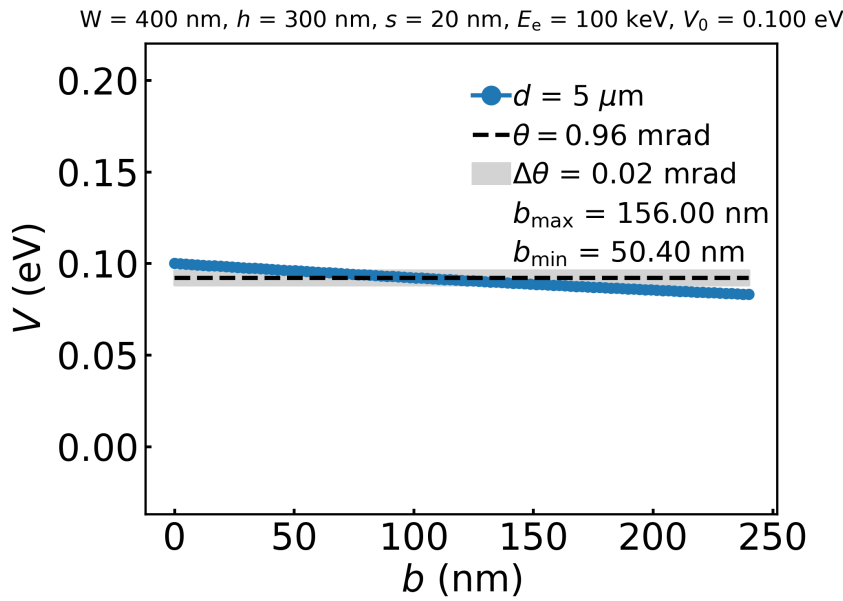
For smaller d , the slop is bigger and $\Delta\theta$ can be bigger



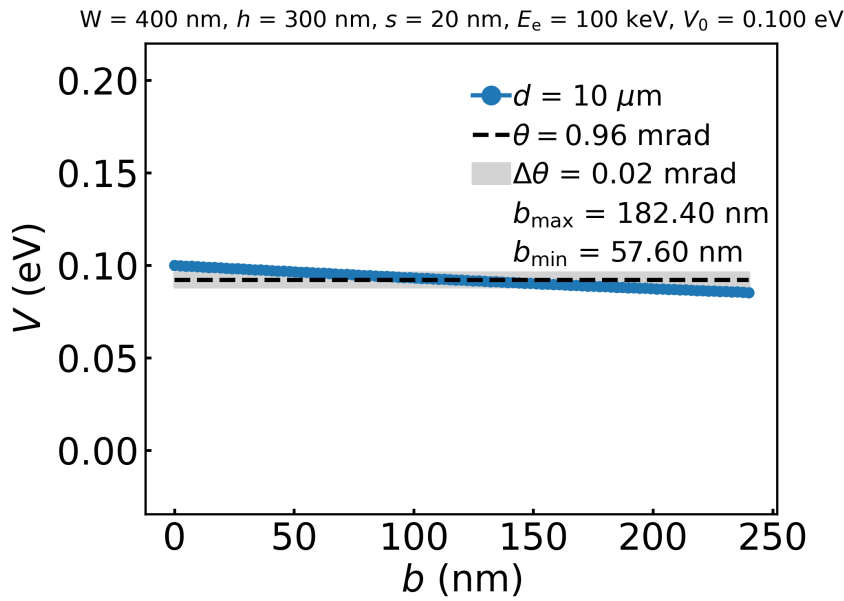
$d = 1 \mu\text{m}$



$d = 5 \mu\text{m}$



$d = 10 \mu\text{m}$



Same as above with different E_e and V_0

